
Status Report:

*Nautatava Navlakha - 160040007

Sarvind Rankawat - 160040057

Arsh Akhtar Ansari - 160040024

*Team leader

Goal and Objectives

Created as an idea to change the traditional approach of pen-paper based survey, this project gives surveyor a way to both decrease the chances of errors and ease the survey taking procedure

No	Major Tasks	Completion Date	Personal Responsible	Weight
	Inception report	Mar. 03	<Group leader>	15%
1	Building the interface for user	Mar. 15	Nautatava	5%
2	Building the API for receiving Data from app	Mar. 22	Arsh	5%
3	Connecting to Maps API	Mar. 29	Sarvind	5%
4	Build interface for surveyor to change questions	Mar. 30	Arsh and Nautatava	10%
5	Building the questionnaire to be asked in survey	Apr. 1	All	15%
6	Take survey from inmates to practically use the app	Apr. 5	All	15%
7	Analyze data to look out for trends	Apr. 7	All	15%
	Final report in ppt and presentation	Apr. 10	All	15%

Interface development

App built on **Java**.

This is the first point where users get a touch of questions. Includes images so that the questions are self explanatory.

types of interfaces:

- User interface to answer questions
- Admin interface to change questions
- Look for the responses given
- Option to export to sheet

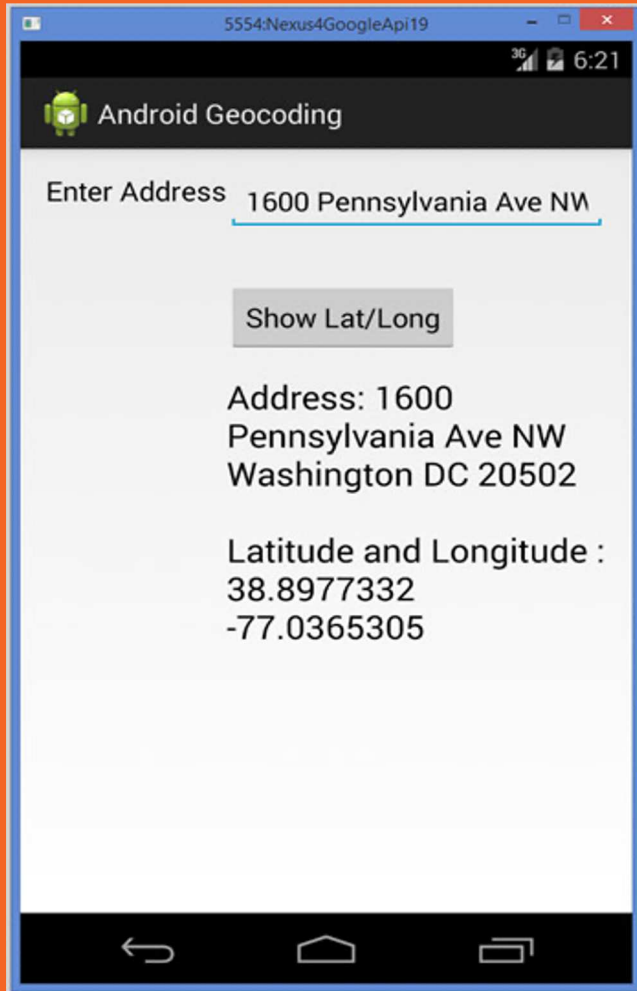


Specifications

App built on android studio and supported on any smartphone with Android 5.0 or higher.

According to Google, almost **85%** devices are supported

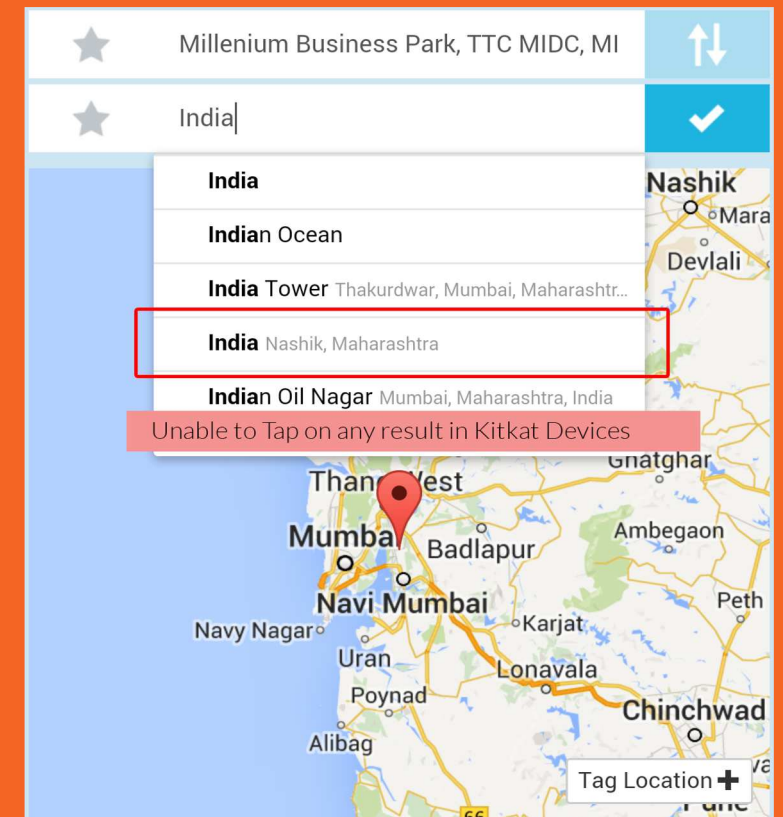
Geocode data extraction



A



- A. Finding textual location using text entered
- B. Calibrating the location data using the map



B

Backend Data Collection

- A. Execution of a django database using different models for each field of database
- B. Execution of APIs for connecting data collection from the app to server
- C. Conversion of data into different file system like excel files for better data analysis

Home › Apis › Surveys

Select survey to change

Action: 0 of 3 selected

☐	NAME	PHONE	GENDER	HOME	WORK
☐	Nautatva Navlakha	8976123450	Male	202, Hostel 7	Civil Building, Academic Section
☐	Sarvind Ran	8789364880	Male	155, Hostel 2	Civil Building, Academic Section
☐	Arsh Akhtar Ansari	9819751710	Male	153, Hostel-2	Civil Building, Academic Section

3 surveys

Admin interface

1. A admin interface where supervisor can add interviewers, set questions and view surveys taken.
2. User logs into his interviewer account to get the questions on the screen. He can then start taking interviews.
3. Extracting responses in form of excel sheet to view and share data offline.

Site administration

APIS

Interviewers

[+ Add](#) [✎ Change](#)

Questionnaires

[+ Add](#) [✎ Change](#)

Surveys

[+ Add](#) [✎ Change](#)



Questionnaire Design

Choose a good questionnaire design to grab the purpose right from the start.

→ **Household**

Get the socioeconomic info

→ **Personal**

Get a detailed house structure in the area

→ **Simple**

Provide a simple unifying message for what is to come

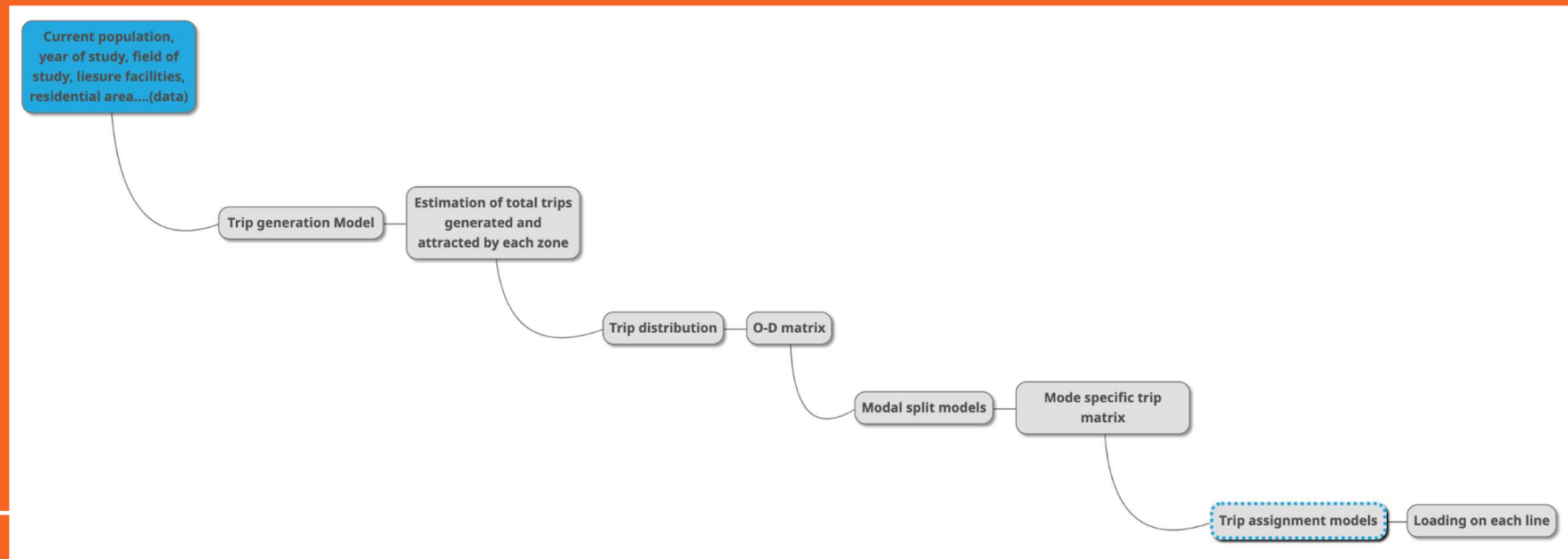
Take survey from inmates

1. The portals being ready, now can be used for practical purpose of taking survey on-field.
2. Will help in checking correctness of app.

☐	NAME	PHONE	GENDER	HOME	WORK
☐	abc	1234567890	Male	Pick a location	Pick another location
1 survey					

Analyze data

1. Analysis is important to know the current transportation situation
2. Our analysis will be making trip generation model





Why not conventional methods?

There are a number of online survey sites such as SurveyMonkey or as simple as google sheets, but....

→ **Flexibility**

When you need to add custom validators to ensure the data provided is correct.

→ **Hefty prices**

The external reports come to us with a hefty price which is way more than what maintaining a self-made app needs.

Summary Report

Expected delivery

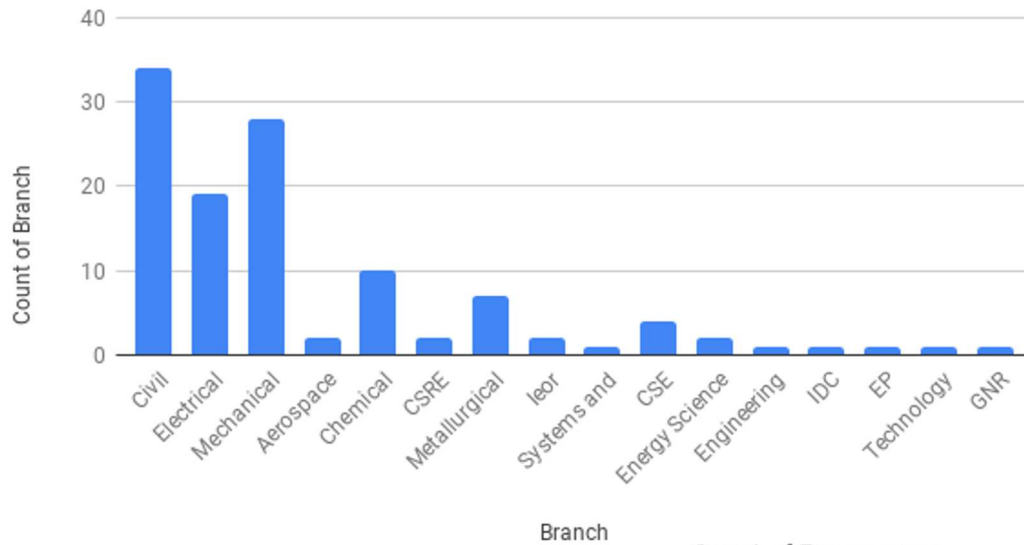
April 15, 2019

For target

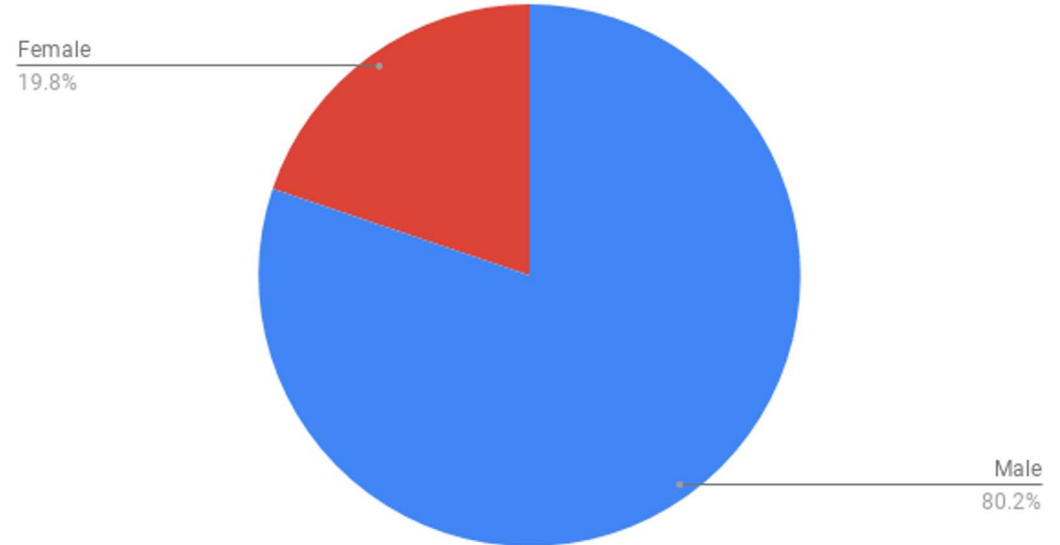
Analysis of trends of students going to various places like Academic Area, SAC, Main gate and Market gate.

Knowing our audience

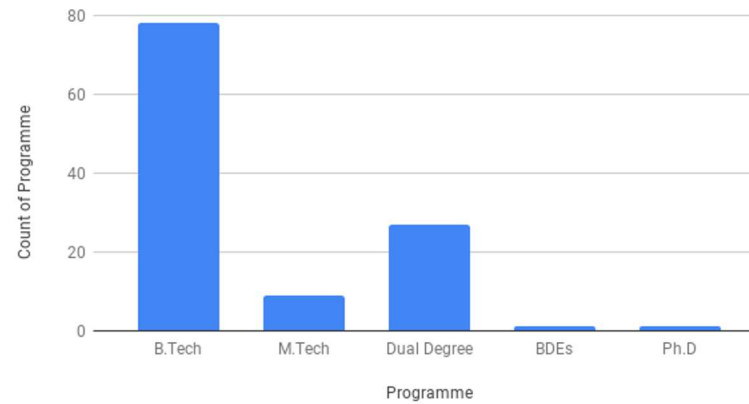
Count of Branch



Count of Gender

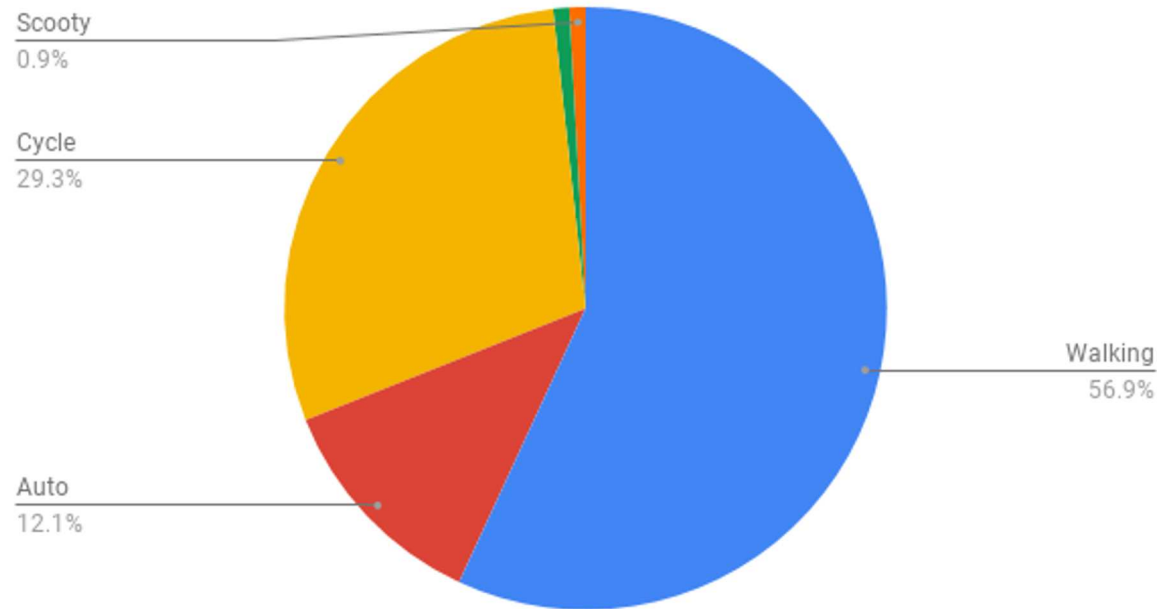


Count of Programme



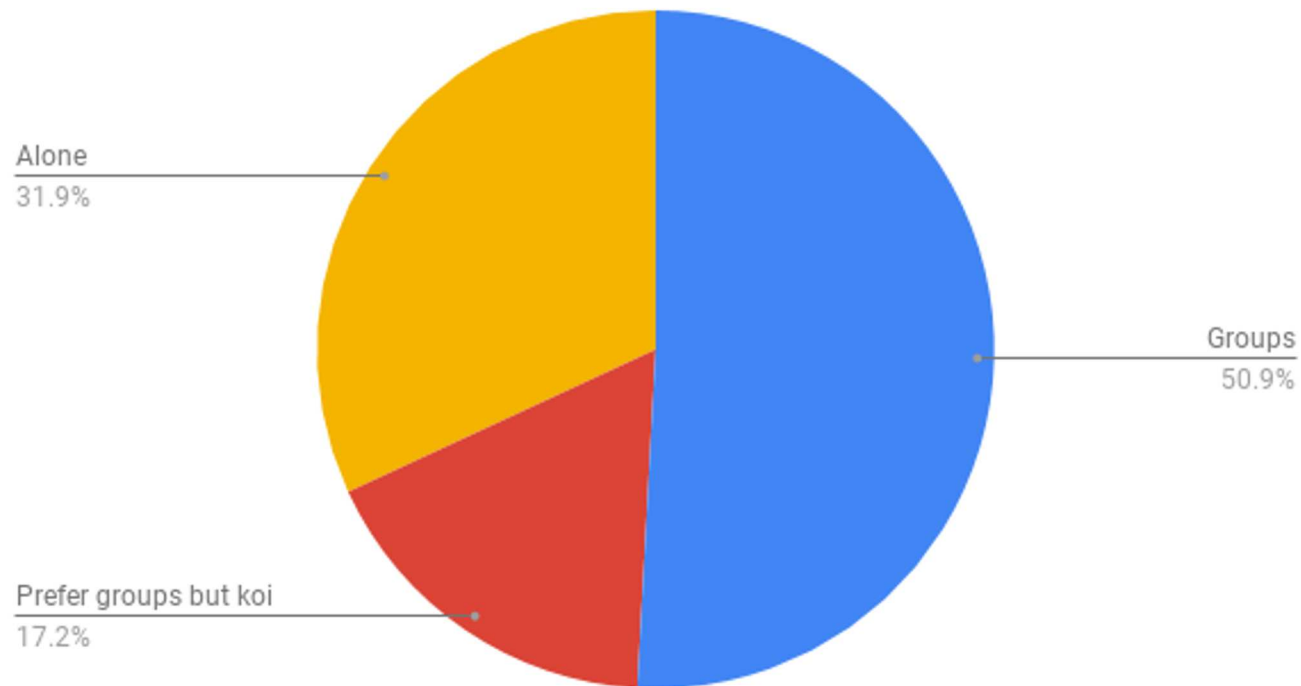
Most common mode of transport

Count of Most frequent mode of transport



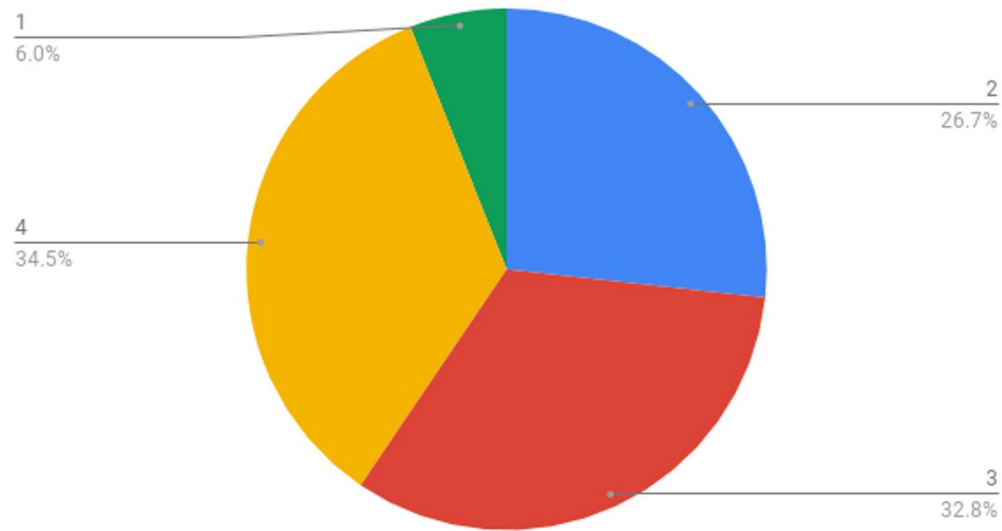
Travel in groups?

Do you prefer travelling alone or in groups?

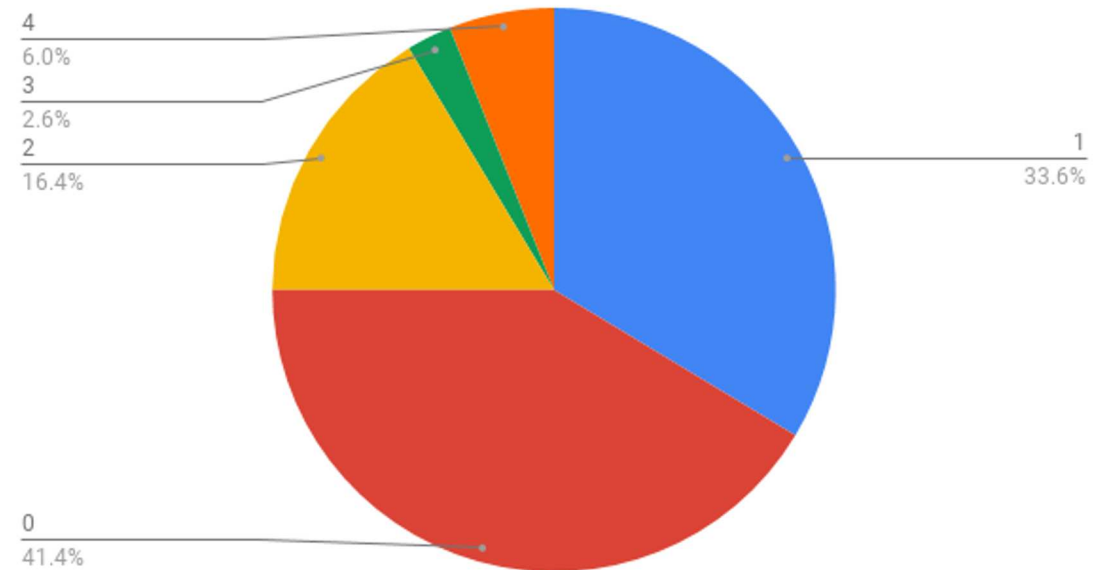


Travel to Academic Area

Number of trips to Academic area on weekdays

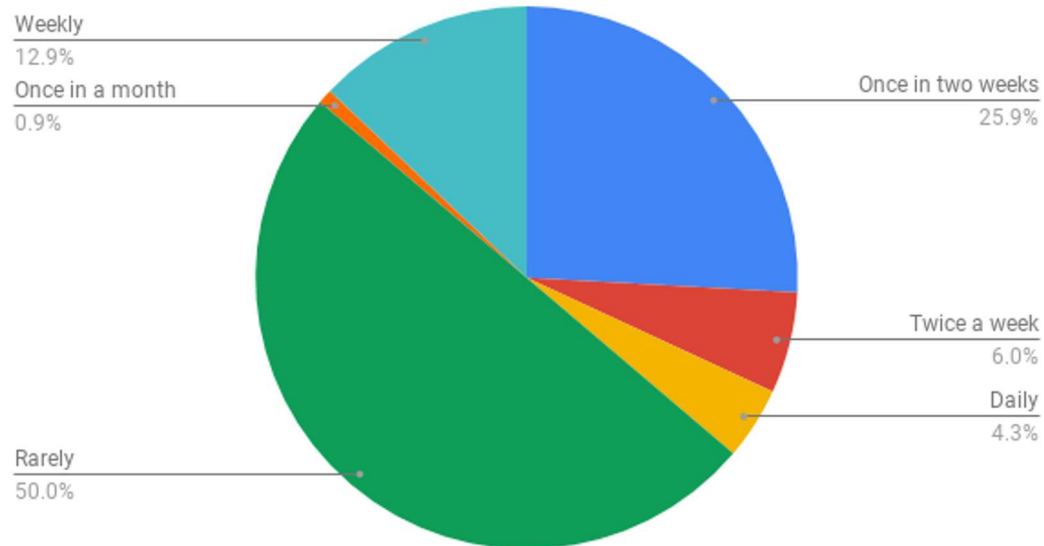


Histogram of Number of trips to Academic area on weekends

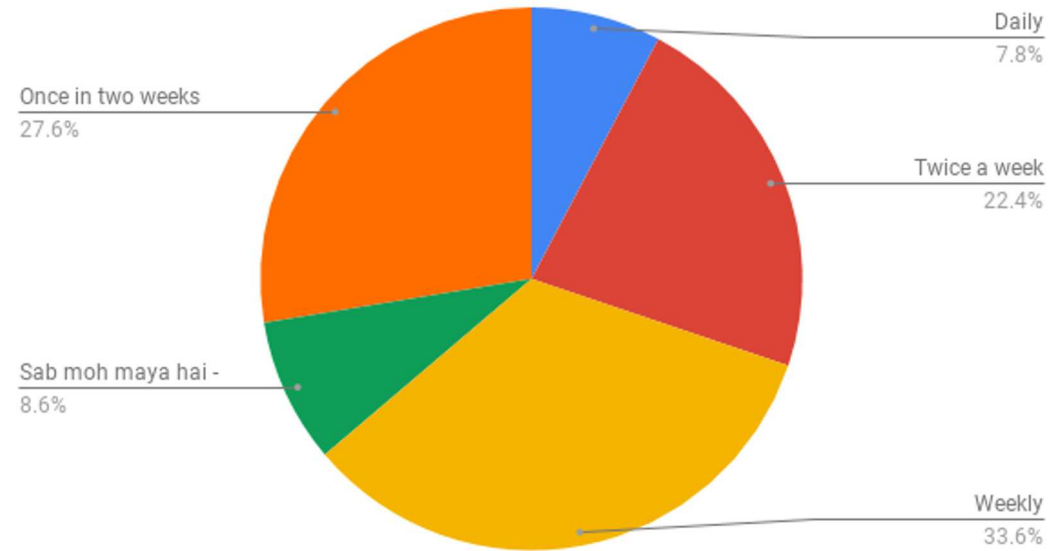


Interaction with the outside world

Count of Frequency of visits to Market Gate

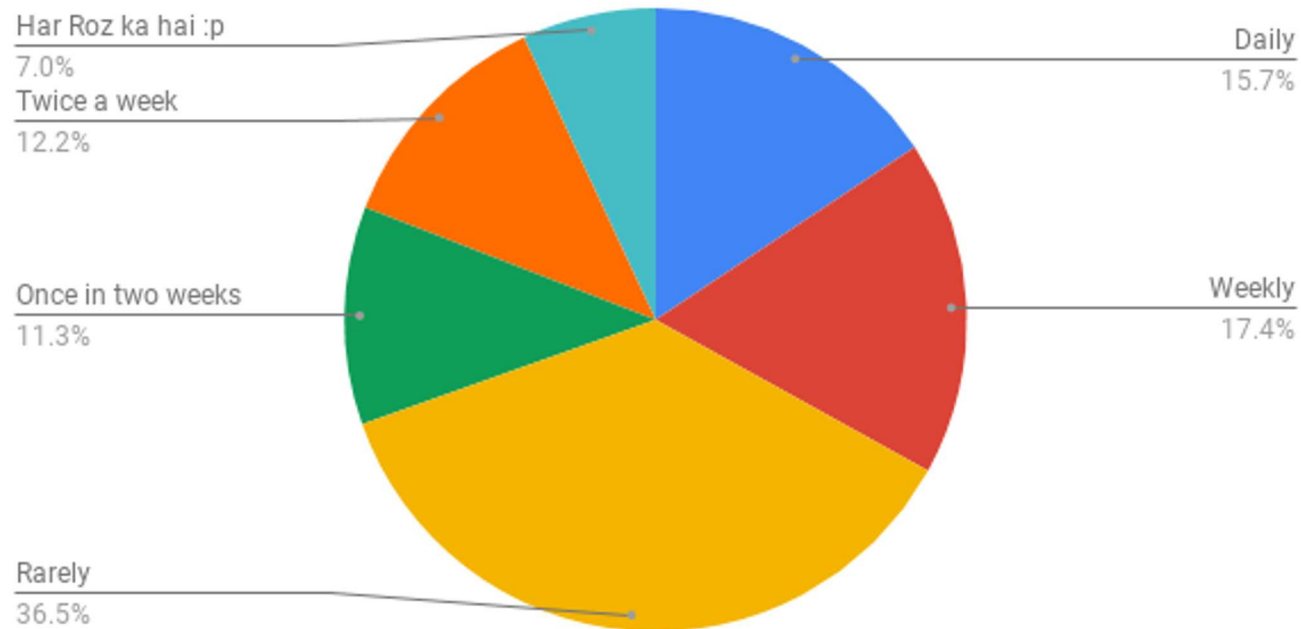


Count of Frequency of visits to Main Gate



Activities other than academics...

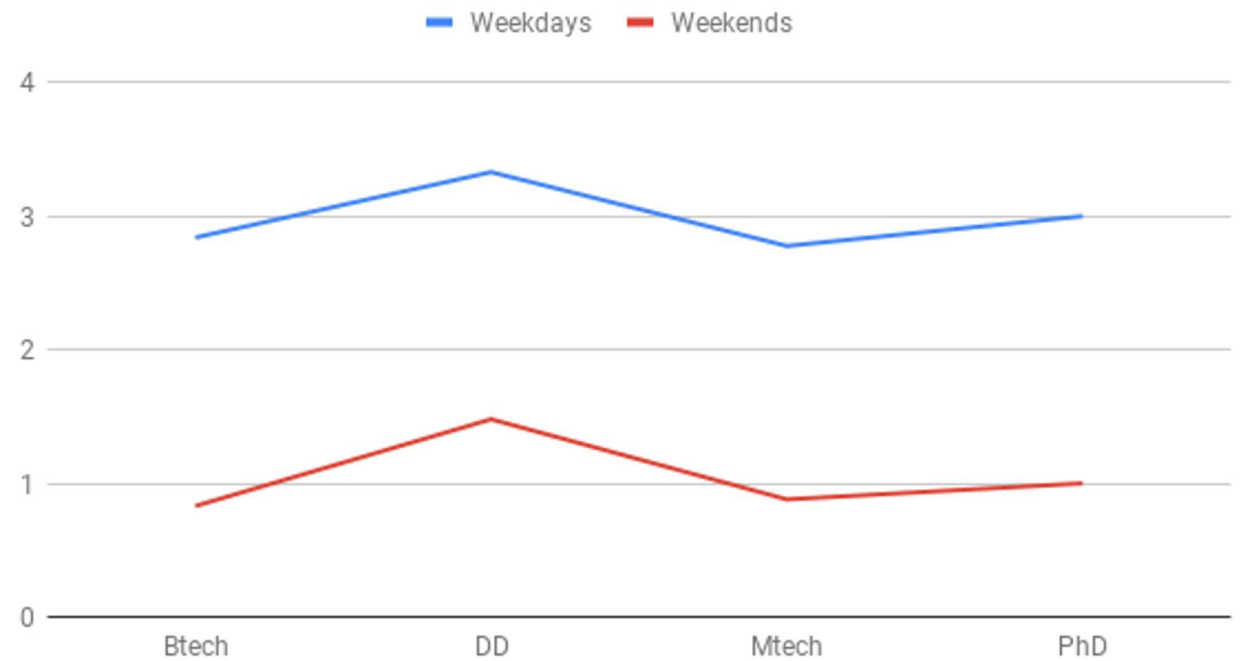
Count of Visit to SAC / Ground / playing services / common gym



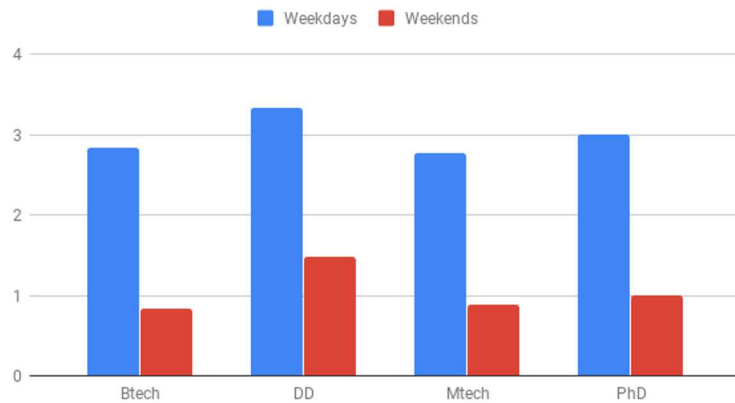
Interesting Observation

	Weekdays	Weekends
Btech	2.84	0.83
DD	3.33	1.48
Mtech	2.777	0.88
PhD	3	1

Weekdays and Weekends



Weekdays and Weekends





Results

The app has an edge in many aspects like flexibility in surveys. Some points in flexibility are as seen -

→ **App Based Data Analytics**

Get graphs based on the data collected directly on the mobile app

→ **Multi survey app**

Rather than using it once and resetting it we can have multiple use for different surveys

Thank You!